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Psychotropic Medications Promote Time-Dependent Reduction of Suicidal Ideation in Mood Disorder: A Prospective Cohort Study

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ABSTRACT

Background: Despite a plethora of research on the topic, there is still no solid evidence that pharmacological treatment actually reduces the risk of suicide in patients with mental illness. In this study, we aimed to assess the effect of psychotropic medications on suicidal ideation in patients with major depressive disorder (MDD) and bipolar disorder (BPD) in two age groups: less than 25 years and 25 years and older.

Methods: We analyzed 312 patients with mood disorders with current suicidal thoughts or recent suicide attempts. We followed the participants from baseline for 6 months and assessed changes in suicidal ideation with Columbia-Suicide Severity Rating Scale (C-SSRS). The effect of psychotropic drug administration on suicidal ideation over time was analyzed using a linear mixed model.

Results: In patients aged 25 years and older with mood disorders, suicidal ideation was more severe when using psychotropic drugs than when not using them. However, suicidal ideation decreased rapidly over time. The time-dependent reduction in suicidal ideation was accelerated when using antidepressants and sedatives/hypnotics in adult MDD, and when using mood stabilizers in adult BPD. However, this effect was not observed in participants aged less than 25 years.

Conclusion: Adequate psychotropic medication may reduce suicidal ideation in patients with

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Disclosure

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Ahn YM. Formal analysis: Yang JH. Investigation: Park CHK, Rhee SJ, Kang DH, Kim MJ, Lee HJ, Lee SY, Shim SH, Moon JJ, Cho SJ, Kim SG, Kim MH, Lee J, Kang WS, Yoo J, Lee WY. Writing - original draft: Yang JH. Writing - review & editing: Ahn YM.

Availability of Data and Material

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

mood disorders aged 25 years and older. Additional research on psychotropic drugs is needed to effectively reduce the risk of suicide among children and adolescents with mood disorders.

Keywords: Suicide; Suicidal Ideation; Psychotropics; Psychiatric Treatment; Antidepressant

INTRODUCTION

According to the World Health Organization, 800,000 people die by suicide each year.¹ In 2021, among Organization for Economic Cooperation and Development member countries, South Korea had the highest suicide mortality rate.² Suicide is a significant medical and social problem in South Korea. According to the 2018 Suicide Survey, 5.3% of the South Korean population have planned suicide and 2.4% have a history of actually attempting suicide.³

However, many of those at high risk of suicide have psychiatric disorders. A psychological autopsy study reported that more than 90% of all suicide deaths involved mental illness.⁴ Among the major psychiatric disorders, mood disorders are a major cause of suicide. Among patients with mood disorders, 30–40% attempt suicide, and 0.5–8.6% die by suicide.⁵ Although there are conflicting prospective studies, in general, schizophrenia and mood disorders are rated as having the highest suicide risk among major psychiatric disorders.^{4,6}

Despite many previous studies, clear evidence on whether treatment for mental illness, especially pharmacological treatment, actually reduces the risk of suicide is still lacking.⁷⁻⁹ Suicide deaths are associated with higher rates of psychotropic use, including antidepressants, sedatives, and hypnotics, compared with the general population.^{10,11} This may be related to the severity of psychiatric symptoms, such as depression, that precedes psychiatric treatment and suicidal death.

Regarding antidepressants, evidence on suicide risk during treatment is not completely conclusive and is sometimes contradictory. In September 2004, the FDA issued a black box warning related to an increased risk of suicidal ideation and behavior in pediatric patients for all antidepressants; in 2007, the warning was expanded to include young adults under the age of 25 years.^{12,13} Data from 4,582 patients in 24 placebo-controlled trials showed that antidepressant use was associated with a slightly increased risk of suicide in pediatric patients.¹⁴ An analysis of 27 trials of selective serotonin reuptake inhibitors (SSRIs), nefazodone, venlafaxine, and mirtazapine, in participants younger than 19 years also found an increased risk with drug use.¹⁵ Of the 15,390 Finnish patients hospitalized after a suicide attempt, the current use of antidepressants increased the risk of suicide attempts but decreased the risk of completed suicide.¹⁶

Taken together, these studies suggest that antidepressants decrease the risk in adults, but not in children and adolescents.¹² Although some researchers have explained this phenomenon by the relatively high prevalence of bipolar disorder (BPD) in young people or by antidepressant-induced activation, whether this phenomenon actually exists is not clear.¹⁷⁻¹⁹ To clearly explain this phenomenon, it is necessary to confirm the effect of antidepressants on suicidality through a prospective study by dividing the participants into BPD and major depressive disorder (MDD) groups, as well as into adult and youth groups.

In addition to antidepressants, other psychotropic drugs have been reported to have various effects on suicide. Large meta-analyses have shown that lithium in mood disorders and

clozapine in schizophrenia prevent suicide.^{20,21} In contrast, anticonvulsants were recently found to be associated with an increased risk of suicide attempt in a 2017 study involving 3,390 manic patients.²²

Some argue that specific drugs do not collectively increase or decrease the risk of suicide but that the risk of suicide can be lowered by using the appropriate drug for the indication according to the patient's accurate diagnosis.²³ For example, the use of antidepressants in patients with MDD, mood stabilizers in patients with BPD, and antipsychotics in patients with schizophrenia may reduce the risk of suicide.²⁴⁻²⁷

Most previous studies dealing with changes in suicide risk according to psychotropic drug treatment were cross-sectional or longitudinal studies with interventions. To construct a risk-prediction model that reflects the various characteristics of patients at high risk of suicide in the real world, it is necessary to verify the effects of psychotropic medication on suicide risk through observational and prospective studies.

In this study, mood disorder patients from an observational, prospective cohort study targeting a high-risk group for suicide, were analyzed using a linear mixed model (LMM). We hypothesized that suicidal ideation would decrease more rapidly over time in adults with mood disorders who are taking appropriate psychotropic medications based on their diagnoses. We also hypothesized that the effects of psychotropic medications on suicidal ideation would differ between groups stratified by age (≥ 25 years and < 25 years).

METHODS

Participants and procedure

K-COMPASS is a cohort study that aims to investigate the course of suicidal ideation and suicide attempts. The original cohort began on September 1, 2015, and the first and second extended cohort have been in progress since July 1, 2019, and July 1, 2022. In this study, only data from the pre-extension and first extended cohorts were analyzed. Both cohorts comprised suicidal ideators and suicide attempters. Suicidal ideators were defined as those with serious suicidal ideation at baseline, and suicide attempters were defined as those with a history of suicide attempts within 1 month before baseline. Participants were enrolled from eight university hospitals across Korea to recruit patients from the outpatient or emergency rooms as well as eight community-based mental health centers associated with the hospitals. Please refer to our previous paper for more information on the results of this study and participant characteristics.²⁸

Among the 1,248 participants enrolled in the cohort, 812 were evaluated as having MDD and 224 as having BPD using the Korean version of the Mini International Neuropsychiatric Interview (K-MINI). Of these, 365 completed the evaluation within 6 months after registration. Among these, 312 people who did not start or discontinue psychotropic medications for six months were analyzed.

Measurements

Socio-demographic and clinical variables

The participants' sociodemographic and clinical characteristics were evaluated by psychiatrists, research nurses, and medical social workers at baseline. Current and previous suicidal thoughts and attempts as well as physical and psychiatric disorders were assessed.

Psychiatric evaluations were performed using the K-MINI and other scales. All evaluators held periodic meetings for education, and the primary investigators at each site participated in these conferences.

Columbia Suicide Severity Rating Scale (C-SSRS)

The Columbia Suicide Severity Rating Scale (C-SSRS) was developed to distinguish the domains of suicidal ideation and suicidal and non-suicidal self-injurious behaviors, differentiate the types of thoughts and behaviors, provide standard definitions, and suggest questions to guide assessment. The scale comprises four parts: severity of suicidal ideation, intensity of suicidal ideation, suicidal behavior, and lethality of the behavior. Previous studies have demonstrated the reliability and effectiveness of the C-SSRS.²⁹ The severity and intensity subscales of the C-SSRS were used to assess suicidal ideation. The severity and intensity subscales range from 0 to 5 and 0 to 30, respectively, with a higher score indicating greater suicidal ideation.

Assessment of use of psychotropic drugs

At each evaluation, the interviewer investigated participants' psychotropic drug use through a structured interview. Participants reported their current psychotropic use and brand or ingredient names for all psychotropic medications they were taking. According to the Anatomical Therapeutic Chemical classification, the psychotropic drugs used by the participants were classified into four categories: antidepressants, mood stabilizers, antipsychotics, and sedatives/hypnotics.³⁰ The researchers determined whether the participants were taking the drug in each category or not.

Statistical analysis

All analyses were performed using SPSS 23 (version 23.0; IBM Corp., Armonk, NY, USA) and were considered statistically significant at $P < 0.05$. Differences in sociodemographic and clinical characteristics between participants were compared using a Student's t -test and χ^2 test. We also used a LMM to evaluate the fixed effects of psychotropic use, time (days), and their interaction on the intensity of suicidal ideation. Both random intercepts and slopes were considered in the LMM analysis. All participants were divided into MDD and BPD groups, and subgroups according to age (≥ 25 years and < 25 years) for analysis. Additionally, we conducted further subgroup analysis to verify the effect of lithium among mood stabilizers on reducing suicidal ideation. We analyzed the fixed effects on the reduction of suicidal ideation over time using a LMM for cases where lithium was taken and for cases where non-lithium mood stabilizers were taken.

Ethics statement

This study was reviewed and approved by the Institutional Review Board of Seoul National University Hospital (H-1505-050-671) and all other participating centers. Informed consent was obtained from all participants upon enrollment.

RESULTS

Comparisons of socio-demographic and clinical characteristics

Table 1 compares the characteristics of the groups aged 25 years and older and those aged less than 25 years. Among the 312 participants with mood disorders, 214 were 25 years and older and 98 were younger than 25 years. Compared to the younger group, the older group were

Table 1. Comparison of baseline characteristics between age groups

Variables	Age ≥ 25 yr (n = 214)	Age < 25 yr (n = 98)	t/χ^2_{df} ^a	P value
Age, yr	48.88 ± 16.90	20.20 ± 2.25	24.27	< 0.001
Sex (male)	77 (36.2)	31 (31.6)	0.60	0.437
Marital status (married)	65 (30.4)	3 (3.1)	29.42	< 0.001
Having a job	117 (54.7)	53 (54.1)	0.01	0.992
Religion (yes)	117 (54.7)	30 (30.6)	15.62	< 0.001
Cohabitant (yes)	144 (67.3)	79 (80.6)	5.85	0.016
Lifetime history of suicide attempt (yes)	160 (75.1)	78 (80.4)	1.05	0.306
Previous diagnosis of physical illness	123 (57.5)	25 (25.8)	26.90	< 0.001
Current psychotropic use	172 (80.4)	92 (93.9)	9.42	0.002
Antidepressants	123 (57.5)	58 (59.2)	0.08	0.777
Mood stabilizers	44 (20.6)	36 (36.7)	9.22	0.002
Antipsychotics	91 (42.5)	62 (63.3)	11.57	< 0.001
Hypnotics/sedatives	130 (60.7)	61 (62.2)	0.06	0.801
K-MINI psychiatric diagnosis			8.43	0.04
Major depressive disorder	165 (77.1)	60 (61.2)		
Bipolar disorder	49 (22.9)	38 (38.8)		
Initial clinical scales				
C-SSRS severity	3.04 ± 1.72	3.61 ± 1.37	-3.15	0.002
C-SSRS intensity	14.66 ± 5.25	16.80 ± 3.73	-3.10	< 0.001
PHQ-9	16.91 ± 6.95	19.580 ± 5.10	-4.07	0.001
BAI	28.36 ± 15.89	30.23 ± 13.15	-0.99	0.322
AUDIT	7.49 ± 10.23	7.21 ± 9.07	0.22	0.823

Values are presented as mean ± standard deviation or number (%). Bold text indicates $P < 0.05$.

K-MINI = Korean version of the Mini International Neuropsychiatric Interview, C-SSRS = Columbia-Suicide Severity Rating Scale, PHQ-9 = Patient Health Questionnaire-9, BAI = Beck Anxiety Inventory, AUDIT = Alcohol Use Disorder Identification Test.

^a χ^2 test for categorical variables and Student's t -test for continuous variables were performed.

more likely to be married (30.4% vs. 3.1%, $P < 0.001$), have a religion (54.7% vs. 30.6%, $P < 0.001$), have been previously diagnosed with a physical illness (57.5% vs. 25.8%, $P < 0.001$), and were less likely to live with a partner (67.3% vs. 80.6%, $P = 0.016$). Other demographic variables except age did not differ between the two groups.

Among the clinical variables, there was no difference between the two groups in the number of lifetime suicide attempts despite the difference in age between the groups. There was no difference in the rate of MDD and BPD between the two groups. The older group were less likely to be currently using psychotropic medications (80.4% vs. 93.9%, $P = 0.002$), especially mood stabilizers (20.6% vs. 36.7%, $P = 0.002$) and antipsychotics (42.5% vs. 63.3%, $P < 0.001$). In clinical scales, C-SSRS severity (3.042 ± 1.715 vs. 3.612 ± 1.367 , $P = 0.002$), intensity (14.659 ± 5.2488 vs. 16.796 ± 3.730 , $P < 0.001$), Patient Health Questionnaire-9 (PHQ-9) (16.906 ± 6.947 vs. 19.575 ± 5.101 , $P = 0.001$) was higher in the younger group, and there were no differences in any other measures.

Table 2 compares the characteristics of patients with MDD and patients with BPD in the group aged 25 years and older. Patients with MDD were older than patients with BPD (51.024 ± 17.301 vs. 41.673 ± 13.273 , $P < 0.001$) and were less likely to have a job (50.3% vs. 69.4%, $P = 0.018$). There were no significant differences in the other demographic variables. Patients with MDD had a lower rate of previous suicide attempts (71.5% vs. 87.5%, $P = 0.024$) and a higher rate of being diagnosed with a physical illness (61.2% vs. 44.9%, $P = 0.043$).

Among patients with MDD aged 25 years and older, 74.5% used psychotropic medications. The rate of use of antidepressants, which are the most commonly used drugs, was also low (58.2%). All patients with BPD in the same age group were taking the prescribed psychotropic medications. The most commonly used medication for BPD was sedative hypnotics, and

Table 2. Comparison of baseline characteristics between major depressive disorder and bipolar disorder (age ≥ 25)

Variables	Major depressive disorder (n = 165)	Bipolar disorder (n = 49)	t/ χ^2 ^a	P value
Age, yr	51.02 $\pm$ 17.30	41.67 $\pm$ 13.27	4.02	< 0.001
Sex (male)	61 (37.0)	16 (33.3)	0.21	0.644
Marital status (married)	48 (29.1)	17 (34.7)	0.56	0.454
Having a job	83 (50.3)	34 (69.4)	5.55	0.018
Religion (yes)	91 (55.2)	26 (53.1)	0.07	0.796
Cohabitant (yes)	106 (64.2)	38 (77.6)	3.04	0.081
Previous suicide attempt (yes)	118 (71.5)	42 (87.5)	5.08	0.024
Previous diagnosis of physical illness	101 (61.2)	22 (44.9)	4.11	0.043
Current Psychotropic use	123 (74.5)	49 (100.0)	15.52	< 0.001
Antidepressants	96 (58.2)	27 (55.1)	0.15	0.702
Mood stabilizers	18 (10.9)	26 (53.1)	41.10	< 0.001
Antipsychotics	56 (33.9)	35 (71.4)	21.73	< 0.001
Hypnotics/sedatives	90 (54.5)	40 (81.6)	11.63	0.001
Initial clinical scales				
C-SSRS severity	2.90 $\pm$ 1.75	3.53 $\pm$ 1.53	-2.46	0.016
C-SSRS intensity	14.30 $\pm$ 5.31	15.88 $\pm$ 4.89	-1.95	0.055
PHQ-9	17.05 $\pm$ 6.37	17.15 $\pm$ 6.60	-0.09	0.926
BAI	27.72 $\pm$ 15.31	31.17 $\pm$ 15.81	-1.36	0.177
AUDIT	5.75 $\pm$ 8.51	7.87 $\pm$ 10.91	-1.40	0.164

Values are presented as mean $\pm$ standard deviation or number (%). Bold text indicates $P < 0.05$.

C-SSRS = Columbia-Suicide Severity Rating Scale, PHQ-9 = Patient Health Questionnaire-9, BAI = Beck Anxiety Inventory, AUDIT = Alcohol Use Disorder Identification Test.

^a χ^2 test for categorical variables and Student's *t*-test for continuous variables were performed.

the rate of use of mood stabilizers, the standard treatment for BPD,³¹ was the lowest among the four classes of psychotropics. There was no difference in the rate of antidepressant use between the adult MDD and BPD groups; however, the usage rates of mood stabilizers (10.9% vs. 53.1%, $P < 0.001$), antipsychotics (33.9% vs. 71.4%, $P < 0.001$), and sedative-hypnotics (54.5% vs. 81.6%, $P = 0.001$) were higher in patients with BPD.

In terms of clinical scales, C-SSRS severity was significantly higher in BPD (2.897 $\pm$ 1.745 vs. 3.531 $\pm$ 1.529, $P = 0.016$), and all other scales were also higher in BPD, but the differences were not statistically significant.

Table 3 compares the characteristics of patients aged less than 25 years with MDD and BPD. Sociodemographic variables did not differ significantly between the two groups.

From a clinical perspective, 90.0% of the patients with MDD and 100% of the patients with BPD were taking psychotropic medications. The most frequently administered medications in young patients with MDD were antidepressants, whereas the most commonly administered medications in patients with BPD were antipsychotics. Antidepressants were administered more frequently to patients with MDD (73.3% vs. 36.8%, $P < 0.001$), whereas mood stabilizers were administered more frequently to patients with BPD (18.3% vs. 65.8%, $P < 0.001$). There were no statistically significant differences in the clinical scale scores between the two patient groups.

Effect of psychotropic medications on suicidal ideation

Table 4 shows the time-dependent effect of psychotropic medications on suicidal ideation in the mood disorder population aged 25 years and older. Among all patients with mood disorders, suicidal ideation was more common when psychotropic drugs were administered ($B = 3.2761$, $P < 0.001$). However, the interaction between psychotropic use and time had a negative fixed effect on suicidal ideation ($B = -0.7160$, $P = 0.049$).

Table 3. Comparison of baseline characteristics between major depressive disorder and bipolar disorder (age < 25)

Variables	Major depressive disorder (n = 60)	Bipolar disorder (n = 38)	t/ χ^2 ^a	P value
Age, yr	20.15 ± 2.40	20.29 ± 2.57	-0.27	0.786
Sex (male)	18 (30.0)	13 (34.2)	0.19	0.662
Marital status (married)	2 (3.3)	1 (2.6)	0.04	0.844
Having a job	32 (53.3)	21 (55.3)	0.04	0.852
Religion (yes)	15 (25.0)	15 (39.5)	2.30	0.130
Cohabitant (yes)	50 (83.3)	29 (76.3)	0.73	0.392
Previous suicide attempt (yes)	46 (76.7)	32 (86.5)	1.40	0.237
Previous diagnosis of physical illness	13 (22.0)	12 (31.6)	1.10	0.294
Current psychotropic use	54 (90.0)	38 (100)	4.05	0.044
Antidepressants	44 (73.3)	14 (36.8)	12.83	< 0.001
Mood stabilizers	11 (18.3)	25 (65.8)	22.55	< 0.001
Antipsychotics	34 (56.7)	28 (73.7)	2.90	0.089
Hypnotics/sedatives	35 (58.3)	26 (68.4)	1.01	0.136
Initial clinical scales				
C-SSRS severity	3.72 ± 1.24	3.45 ± 1.55	0.95	0.345
C-SSRS intensity	16.97 ± 3.34	16.53 ± 5.77	0.43	0.671
PHQ-9	18.90 ± 5.22	17.22 ± 4.89	1.57	0.120
BAI	30.83 ± 13.45	27.81 ± 13.85	1.06	0.290
AUDIT	6.92 ± 8.37	6.56 ± 8.02	0.21	0.837

Values are presented as mean ± standard deviation or number (%). Bold text indicates $P < 0.05$.

C-SSRS = Columbia-Suicide Severity Rating Scale, PHQ-9 = Patient Health Questionnaire-9, BAI = Beck Anxiety Inventory, AUDIT = Alcohol Use Disorder Identification Test.

^a χ^2 test for categorical variables and Student's t -test for continuous variables were performed.

When analyzing only patients with MDD aged 25 years and older, a positive fixed effect of drug use and a negative fixed effect of the interaction between drug use and time were also confirmed for all psychotropic drugs (for use, $B = 3.0449$, $P = 0.002$; for interaction, $B = -0.0140$, $P = 0.048$), antidepressants (for use, $B = 3.8738$, $P < 0.001$; for interaction,

Table 4. Effect of psychiatric medications on the intensity of suicidal ideation of mood disorder patients over time (age ≥ 25)

Variables	All mood disorders (n = 214)			MDD (n = 165)			BPD (n = 49)		
	B	SE	P value	B	SE	P value	B	SE	P value
All psychotropics									
Use	3.2761	0.8864	< 0.001	3.0449	0.9556	0.002	0 ^b	0 ^a	0 ^b
Time (days)	0.0007	0.0036	0.845	-0.0051	0.0031	0.100	-0.0045	0.0045	0.331
Use × Time	-0.7160	0.0089	0.049	-0.0140	0.0070	0.048	0 ^b	0 ^a	0 ^b
Antidepressant									
Use	2.3695	0.6517	< 0.001	3.8738	0.6211	< 0.001	1.3920	1.2781	0.281
Time (days)	-0.0063	0.0042	0.880	0.0023	0.0036	0.530	-0.0051	0.0059	0.392
Use × Time	-0.0098	0.0067	0.142	-0.0147	0.0058	0.012	> -0.0001	0.0092	0.998
Mood stabilizers									
Use	0.1850	0.8247	0.823	0.3755	1.1908	0.753	1.7206	1.2616	0.180
Time (days)	-0.0052	0.0072	0.472	-0.0143	0.0084	0.089	-0.0242	0.0063	0.001
Use × Time	-0.0112	0.0081	0.167	-0.0110	0.0089	0.220	-0.0264	0.0089	0.008
Antipsychotics									
Use	-0.3364	0.6856	0.624	0.0415	0.6686	0.951	-1.0429	1.4041	0.461
Time (days)	0.0027	0.0048	0.572	0.0067	0.0048	0.162	-0.0041	0.0050	0.421
Use × Time	-0.0019	0.0066	0.774	-0.0030	0.0060	0.617	0.0007	0.0107	0.949
Hypnotics & sedatives									
Use	2.0466	0.6797	0.003	2.7682	0.6274	< 0.001	-1.1305	1.7246	0.515
Time (days)	-0.0067	0.0041	0.845	0.0014	0.0037	0.701	-0.0044	0.0050	0.381
Use × Time	-0.0121	0.0068	0.077	-0.0163	0.0058	0.005	0.0002	0.0115	0.987

Bold text indicates $P < 0.05$.

MDD = major depressive disorder, BPD = bipolar disorder, SE = standard error.

^aAdjusted for age and sex.

^bBecause all bipolar disorder patients were taking psychotropics, the effect of psychotropic use could not be calculated.

Table 5. Effect of psychiatric medications on the intensity suicidal ideation of mood disorder patients over time (age < 25)

Variables	All mood disorders (n = 98)			MDD (n = 60)			BPD (n = 38)		
	B	SE	P value	B	SE	P value	B	SE	P value
All psychotropics									
Use	1.0039	1.7036	0.556	0.4425	1.3729	0.748	0 ^a	0 ^a	0 ^a
Time (days)	0.0010	0.0039	0.792	-0.0064	0.0031	0.043	-0.0053	0.0048	0.279
Use × Time	0.0005	0.0165	0.974	-0.0015	0.0102	0.882	0 ^a	0 ^a	0 ^a
Antidepressant									
Use	0.1114	0.8183	0.892	1.2111	0.9342	0.199	0.1778	1.4706	0.904
Time (days)	0.0005	0.0051	0.926	0.0087	0.0043	0.045	-0.0110	0.0081	0.183
Use × Time	-0.0013	0.0077	0.869	0.0011	0.0080	0.888	-0.0088	0.1000	0.384
Mood stabilizers									
Use	1.5547	0.8244	0.060	2.0533	1.0501	0.055	0.6195	1.4735	0.676
Time (days)	-0.0012	0.0060	0.847	-0.0007	0.0078	0.927	-0.0023	0.0055	0.682
Use × Time	-0.0032	0.0078	0.679	-0.0108	0.0088	0.222	0.0115	0.0102	0.267
Antipsychotics									
Use	0.2954	0.8465	0.727	0.3250	0.8406	0.700	-1.3248	1.6614	0.430
Time (days)	0.0039	0.0047	0.414	0.1000	0.0047	0.034	-0.0015	0.0055	0.790
Use × Time	0.0081	0.0079	0.312	0.0053	0.0073	0.474	0.0167	0.0106	0.127
Hypnotics & sedatives									
Use	-0.4759	0.8425	0.573	-0.3332	0.8545	0.698	-0.2814	1.5084	0.853
Time (days)	0.0003	0.0048	0.952	0.0101	0.0047	0.035	-0.0083	0.0057	0.154
Use × Time	-0.0018	0.0079	0.816	0.0051	0.0074	0.490	-0.0106	0.0100	0.298

Bold text indicates $P < 0.05$.

MDD = major depressive disorder, BPD = bipolar disorder, SE = standard error.

^aAdjusted for age and sex.

^aBecause all bipolar disorder patients were taking psychotropics, the effect of psychotropic use could not be calculated.

$B = -0.0147$, $P = 0.012$), and sedative-hypnotics (for use, $B = 2.7682$, $P < 0.001$; for interaction, $B = -0.0163$, $P = 0.005$). In other words, patients with MDD aged 25 years and older had more severe suicidal thoughts while using any psychotropic medication, antidepressants, or sedative-hypnotic drugs, but suicidal ideation tended to improve over time while taking medicine. In addition, when analyzing patients with BPD aged 25 years and older, a negative fixed effect of the interaction between mood stabilizer use and time was observed ($B = -0.0264$, $P = 0.008$), and through this, the effect of mood stabilizer use on promoting time-dependent reduction of suicidal ideation was confirmed.

However, the effects of these medications were not replicated in participants aged less than 25 years. **Table 5** shows the effect of psychotropic use on suicidal ideation over time in the population aged less than 25 years with mood disorders. The protective effect of medications on suicidal ideation observed in the older group was not observed in the younger group.

Meanwhile, additional subgroup analysis showed that non-lithium mood stabilizers had a positive effect on reducing suicidal ideation over time in patients aged 25 years and older with mood disorders. However, this effect was not observed when lithium was used or when the effect was analyzed specifically for MDD or BPD. In patients under 25 years old, neither lithium nor non-lithium mood stabilizers were effective in reducing suicidal ideation (**Supplementary Tables 1 and 2**).

DISCUSSION

This study showed that psychotropic medication had a positive effect on the reduction of suicidal ideation over time in patients aged 25 years and older. While psychotropics were being administered, suicidal ideation tended to decrease over time compared to when

psychotropic drugs were not being administered. Antidepressants, sedatives, and hypnotics had a positive effect on the time-dependent reduction of suicidal ideation in patients with MDD. Previous studies have suggested that mood stabilizers may promote a decline in suicidal ideation over time in patients with BPD. However, these effects were not observed in patients aged less than 25 years in the current study.

Comparison of patients aged 25 years and older and under 25 showed that the degree of suicidal ideation and depression was more severe in patients in the younger age group. This may be partially attributed to atypical depression, which is more common in young depressed patients. Atypical depression is known to have more severe symptoms, a higher risk of suicide, and a high correlation with BPD.^{32,33} Among patients with MDD who participated in this study, when comparing the different age groups in an additional analysis, C-SSRS severity (2.897 vs. 3.717, $P < 0.001$), C-SSRS intensity (14.297 vs. 16.967, $P < 0.001$), and PHQ-9 scores (17.048 vs. 18.900, $P = 0.045$) were all significantly higher in the younger age group, but this difference was not observed in patients with BPD. Previous studies have suggested that the severity of depression is not related to differences in the effectiveness of medication treatments.^{34,35} Therefore, we supposed that the differences in severity between the two groups did not cause the differences in medication effects observed in this study.

Meanwhile, the prevalence of BPD was significantly higher in the group under 25 years old. This difference could have resulted in the underestimation of the effectiveness of antidepressants or the overestimation of the effectiveness of mood stabilizers in this group. However, even when the group under 25 years old was divided and analyzed separately for MDD and BPD, no type of psychiatric medication was effective in reducing suicidal ideation. Therefore, the difference in the proportions of MDD and BPD according to age is not expected to have had a decisive impact on the results of this study.

In the older patient group, patients with MDD were older and had a higher rate of physical illness than patients with BPD. This is consistent with the knowledge that BPD develops at a younger age.^{36,37} Despite their younger age, patients with BPD were more likely to attempt suicide and have more severe suicidal thoughts. This is also consistent with a previous paper reporting that the risk of suicide in BPD is higher than in MDD.^{38,39}

The most common drugs taken by patients with MDD aged 25 years and older were antidepressants, which is considered an appropriate prescription. However, only half of the patients were taking antidepressants. In contrast, although patients with BPD in the same age group were prescribed psychotropic medications, the rate mood stabilizer administration, known as the standard treatment for BPD, was the lowest among the drug classes. In summary, the rate of adult MDD receiving psychopharmacological treatment was low, and the rate of adult patients with BPD receiving treatment was high, but the rate of appropriate medication administration was low.

On the other hand, in the group of patients aged less than 25 years, not only patients with BPD but also those with MDD were taking psychotropic drugs at a high rate. Compared with patients aged 25 years and older, the proportion of patients taking antidepressants for MDD and mood stabilizers for BPD, which are generally considered appropriate prescriptions, was higher.

According to the 2021 Korea Mental Health Survey conducted by the Ministry of Health and Welfare, the prevalence of mental disorders among Korean citizens was 27.8%; however, the

mental health service utilization rate among patients diagnosed with mental disorders was only 4.4%.⁴⁰ In this study, patients were recruited primarily from hospitals, and although patients with a high severity of suicidal ideation were recruited, a significant number of patients were not taking psychotropic medications. According to a 2019 US Center for Disease Control (CDC) survey, younger people tend to more actively seek treatment for depression than older people,⁴¹ and a similar trend of treatment inadequacy may exist in older adult patients in Korea with regard to the results of our study. Korea, a global high-risk suicide country, is particularly renowned for its high suicide rate among the elderly,⁴² and it is possible that inappropriate psychotropic drug treatment in older adults in this study may be contributing to the suicide rate.

This study found that psychotropic drug use in mood disorder patients aged 25 years and older promoted the time-dependent decline of suicidal ideation. This is consistent with the results of previous studies on patients with mood disorder.⁴³ In our study, the intensity of suicidal ideation was more severe during the period of psychotropic use in the older age group, which is consistent with the results of previous studies that showed a higher risk of suicide during the period of psychotropic drug use.^{10,44,45} According to the common view, psychotropic drugs require a certain amount of time to demonstrate efficacy through changes in neural circuits.⁴⁶ Suicidal ideation before these effects appeared to be higher during the period of taking psychotropic drugs. In addition, there is a possibility that patients with severe psychiatric symptoms will more actively receive psychiatric drug treatment.

Similarly, in patients with MDD aged 25 years and older, the intensity of suicidal ideation was more severe during the period of taking antidepressants but decreased faster when antidepressants were being taken. This is consistent with the results of a 2021 cohort study of adult patients using SSRIs, who showed an increased risk of suicide within the first 30 days of drug use and a decreased risk thereafter.⁴⁷

Mood stabilizers also had a beneficial effect on changes in suicidal ideation over time in patients with BPD aged 25 years and older. This may be because mood stabilizers effectively control symptoms such as depression and anxiety in patients with BPD.⁴⁸ Moreover, lithium has a powerful suicide prevention function itself.⁴⁹ However, this effect was not observed in patients aged less than 25 years. Prior reports have shown that mood stabilizers such as lithium, valproate, and lamotrigine have not been clearly demonstrated as effective in pediatric patients, and our results appear to be consistent with these previous studies.⁵⁰

In the subgroup analysis, the effect of non-lithium mood stabilizers in reducing suicidal ideation was confirmed in patients aged 25 years and older with mood disorders. However, this effect was not observed in BPD, and the reduction in suicidal ideation was not confirmed for lithium either. Ultimately, the reduction in suicidal ideation observed in this study for mood stabilizers, including both lithium and non-lithium medications, may be more reasonably interpreted as a result of symptom improvement in BPD patients aged 25 years and older through the use of mood stabilizers, rather than the direct effect of any specific individual medication.

In the case of antipsychotics, no association was found between the medication use and changes in suicidal ideation over time. Antipsychotics, most notably clozapine, have been reported to have a positive effect on reducing suicidality in schizophrenia.⁵¹⁻⁵³ However, antipsychotics have not been reported to reduce suicidal ideation in patients with mood

disorders. Rather, there is an argument that antipsychotics may trigger suicidality.⁵⁴ The results of this study on antipsychotics can be regarded as a combination of the various findings on the effects of antipsychotics on suicidal tendencies.

Patients with MDD aged 25 years and older had more suicidal ideation during the period of use of hypnotics or sedatives; however, suicidal ideation declined more rapidly over time. This may be related to the beneficial effects of hypnotics/sedatives on anxiety symptoms, which are directly linked to suicidal ideation.⁵⁵⁻⁵⁷ The risk of suicide can be mitigated by the timely use of hypnotics/sedatives with immediate effect, especially when the patient's symptoms are severe.

In contrast, no medications were effective in improving suicidal ideation in patients aged less than 25 years. According to previous studies, the effectiveness of these medications in preventing suicide among children and adolescents remains unclear.⁵⁸⁻⁶⁰ The results of the present study appear to reflect the pessimistic results of previous studies.

In patients with MDD aged less than 25 years, antidepressants did not show the protective effects shown in those aged 25 years and older. This result partially supports previous studies that have shown that antidepressants lack protective effects or have detrimental effects in terms of suicide in children with MDD.^{61,62} The degree of improvement of antidepressants on depressive symptoms observed in adults has not been demonstrated in children and adolescents with depression. There have also been reports that drugs such as venlafaxine and paroxetine worsen symptoms, which are thought to be due to neurodevelopmental mechanisms that include strong changes in hormone receptors in children and adolescents.⁶³⁻⁶⁵ As seen in this study, in patients with MDD under the age of 25, if motivation, which is known to respond relatively well to pharmacotherapy among depressive symptoms, is restored without a clear improvement in suicidal thoughts due to antidepressants, the patient is likely to have an increased risk of suicide as they may actually put suicidal ideations into action.^{66,67}

This study is the first prospective cohort study of high-risk suicide participants conducted in Korea. We recruited participants from hospitals and communities in various regions of Korea to minimize selection bias. However, this study did have some limitations. First, a large number of patients dropped out during follow-up, which may have led to inevitable selection bias. Second, we assessed suicidal ideation rather than the suicide itself, which may not directly reflect the actual risk of suicide. However, in this study, we used the C-SSRS, which can quantify suicidal ideation and accurately predict suicide risk.⁶⁸ Third, the status of medication use was assessed entirely based on the patient's self-report, which may differ from the patient's actual medication.

In conclusion, this study showed that psychotropic use has a beneficial effect on suicidal ideation over time in patients with mood disorders aged 25 years and older. We also showed that in patients aged 25 years and older, the use of appropriate psychotropic drugs such as antidepressants, sedatives, and hypnotics for MDD and mood stabilizers for BPD could improve the time-dependent reduction of suicidal ideation, whereas in patients aged less than 25, the drugs did not show any effects. The results of this study suggest that more proactive psychotropic treatment appropriate for a patient's diagnosis is needed to prevent suicide among adults with mood disorders in South Korea. Additional drug research is required to reduce the risk of suicide in children and adolescents with mood disorders.

SUPPLEMENTARY MATERIALS

Supplementary Table 1

The effect of lithium and non-lithium mood stabilizers on time-dependent changes in suicidal ideation in patients with mood disorders (age ≥ 25)

Supplementary Table 2

The effect of lithium and non-lithium mood stabilizers on time-dependent changes in suicidal ideation in patients with mood disorders (age < 25)

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